

Wide awake

Surgery in the past was **BRUTAL**. Patients were left wide awake or groggy with alcohol as cuts were made in their bodies. The **pain or shock** often resulted in death.

Then, in 1846, American dentist **William Morton** discovered that he could use the chemical ether to make a person unconscious before surgery.

Rubber hose carries air and ether fumes from the jar to the mouthpiece to be inhaled.

Small pieces of sponge soaked in ether give off fumes.



Chloroform

Discovered in 1831, chloroform was first used as an anesthetic by Scottish physician **James Young Simpson** in 1847. He used it to provide **PAIN RELIEF** for women, including Queen Victoria, during childbirth. However, chloroform had **dangerous side effects**. Some devices, such as this Dubois machine (left), tried to make it safer by mixing it with air.



Anesthetics

Putting patients to **SLEEP** was the first step toward modern surgical techniques

How it changed the world
Before anesthetics, surgery was quick, brutal, and often deadly. Now, operations can be carried out easily and safely.

Modern techniques

Today, anesthetics can be **local** (numbing a body part such as a foot) or **general** (making a patient unconscious). Amylocaine, the first human-made local anesthetic, was developed by French chemist **ERNEST FOURNEAU** in 1903.

Some general anesthetics are **administered using an injection** and work in less than 30 seconds. Others are given as a carefully controlled dose to be inhaled.

